MEMORANDUM

TO: Michael Kantor, Lead External Audit Partner

FROM: Laveto Lab Systems Architecture & Compliance

SUBJECT: Technical Remediation Dossier: AW-1 Core Ledger Hardening (v2.0.1)

DATE: October 20, 2026

CLASSIFICATION: RESTRICTED — AUDIT PAPERS

STATUTORY BASIS: Bank of Botswana EPS Act, Sec. 14

Executive Overview

This technical dossier details the complete remediation of architectural technical debt, cryptographic deficiencies, and simulation leakage identified during the external compliance audit of the AW-1 Ledger. As of October 20, 2026, all internal code modifications, non-blocking schema upgrades, optimistic concurrency controls, and multi-currency normalizations have been deployed and verified against the live database environment (~/aw1-breaker). All simulated Hedera Consensus Service (HCS) records have been strictly quarantined and tagged under an explicit sandbox protocol, ensuring zero commingling with production consensus assertions.

Remediation Status Matrix

Remediation Item	Target	Intervention	Verification Result	Status
TSK-01: Non-Blocking Schema	ledger_entries	Added parent_hash & node_telemetry via ALGORITHM=INSTANT.	Zero locking; verified non-blocking column availability.	**COMPLETED**
TSK-02: Cryptographic CAS Loop	release_dossier.py	CAS retry loop with jittered exponential backoff & row locks.	Eliminated write-lock deadlocks; parent tip chained.	**COMPLETED**
TSK-03: Historical Lineage Backfill	ledger_entries (#31-#43)	Calculated SHA-256 parent lineage anchored at Genesis Root.	100% cryptographic continuity; 0 broken links.	**COMPLETED**
TSK-04: HCS Sandbox Monotonicity	aw1_hcs_worker.py, hcs_config.json	Enforced monotonic sequence ($S_{i+1} = S_i + 1 >= 512,111$); explicit mode tagging.	Purged mock strings; quarantined from mainnet.	**COMPLETED**
TSK-05: Multi-Currency Normalization	accounts, v_accounts_normalized	Migrated to balance_nominal; rebuilt view with strict NULL separation.	Commingling mathematically prohibited; 376.19% USD solvency.	**COMPLETED**
TSK-06: Hedera Topic Provisioning	External Portal	Provisioning dedicated Testnet/Mainnet Topic ID & custodial key.	Awaiting external operator credentials (HOL path).	**BLOCKED**
TSK-07: Depository Escrow Letter	Custodian Bank	Securing bilateral depository confirmation letter under BoB EPS Sec. 14.	Awaiting commercial bank fiduciary validation.	**BLOCKED**

Detailed Engineering Deliverables

1. Cryptographic Lineage & Concurrency Hardening (TSK-01, TSK-02, TSK-03)

- **Non-Blocking Schema Migration:** ledger_entries was updated via ALGORITHM=INSTANT to introduce parent_hash VARCHAR(64) and node_telemetry JSON without metadata locking.
- **Optimistic CAS Engine:** release_dossier.py implements a 5-attempt retry loop with exponential jitter ($8.0\text{ms} * 2^{\text{attempt}} + U(1.0, 4.0)\text{ms}$), handling deadlocks (1213) and lock timeouts (1205).
- **Tip Resolution:** Atomically resolves latest chain tip via SELECT entry_id, dossier_hash FROM ledger_entries ORDER BY entry_id DESC LIMIT 1 FOR UPDATE;.
- **Lineage Backfill:** Entries #31-#43 were backfilled under baseline note AW1-NOTE-AUDIT-BASELINE-20261019. Entry #31 serves as Genesis Root (0000...00), linking unbroken to tip Entry #43 (0624a8c0...).

```
[Genesis #31: 455847f3...] -> [Entry #32: 9ef34d39...] -> ... -> [Entry #43: 0624a8c0...] -> [Next Release: parent=0624a8c0...]
```

2. Consensus Sandbox Isolation & Monotonicity (TSK-04)

- **Monotonic Counter:** Replaced modulo timestamping with state-backed incrementor initialized from COALESCE(MAX(hcs_sequence_number), 512110), guaranteeing strictly increasing sequence receipts.
- **Purged Mock Values:** Eliminated 0.0.YOUR_TOPIC_ID and double prefix 0.0.0.0.. Identifier space is normalized to 0.0.SIMULATED_SANDBOX_TOPIC and 0.0.SIMULATED_OPERATOR.
- **Quarantine Envelope:** Payloads in aw1_hcs_anchors are permanently stamped with "mode": "SIMULATED_SANDBOX" to prevent conflation with mainnet receipts.

3. Multi-Currency Normalization & Zero-Commingling View (TSK-05)

- **Schema Separation:** Added balance_nominal DECIMAL(18, 2) to accounts, eliminating reliance on balance_bwp for foreign assets.

- **Zero-Commingling View:** Rebuilt v_accounts_normalized such that foreign currencies (USD, ZAR) return NULL for domestic_bwp_balance, mathematically preventing arithmetic pooling.

```
CREATE OR REPLACE VIEW v_accounts_normalized AS
SELECT account_id, account_name, balance_nominal, balance_nominal AS native_balance, currency,
       (CASE WHEN currency = 'BWP' THEN balance_nominal ELSE NULL END) AS domestic_bwp_balance,
       (CASE WHEN account_id IN ('STATUTORY_RESERVE_VAULT','FX_CLEARING_BWP','STATUTORY_RESERVE_USD','STATUTORY_RESERVE_ZAR')
                  OR account_name LIKE '%Reserve%' THEN 'RESERVE_ESCROW'
             WHEN account_id LIKE 'BW-AUDIT%' THEN 'TEST_STAGING' ELSE 'CLIENT_LIABILITY' END) AS regulatory_classification,
       updated_at FROM accounts;
```

Ledger State & Solvency Verification Baseline

Ledger Partition Summary

Partition Scope	Entries	Count	Volume	Verification Status	Operational Handling
Historical Production	#01-#15	15	P 3,142,880.00	Verified Baseline	Active Ledger History
Quarantine Partition	#16-#26	11	P 130,000.00	Quarantined	Segregated; 0 ISO 20022 Emitted
Staging Reconciliation	#27-#30	4	P 0.00	Verified Zero-Sum	Reconciled Audit Artifacts
Active Chained Ledger	#31-#43	13	P 1,024,583.50	Fully Chained	Live Tip: Parent Hash Linked

Multi-Currency Silo Solvency Matrix

Silo	Statutory Escrow Account	Escrow Balance	Client Liabilities	Ratio	Compliance Status
BWP	STATUTORY_RESERVE_VAULT + FX_CLEARING	P 517,463.50	P 482,536.50	**107.24%**	Compliant (EPS Sec. 14)
USD	STATUTORY_RESERVE_USD	$15,800.00	$4,200.00	**376.19%**	Solvent (+$11,600.00 Surplus)
ZAR	STATUTORY_RESERVE_ZAR	R 1,986,550.00	R 13,450.00	**14,769.88%**	Fully Solvent

External Production Dependencies (TSK-06 & TSK-07)

1. **TSK-06 (Hedera Network Topic Provisioning):** Requires dedicated Hedera operator account and consensus topic ID provisioned via Hedera Portal. When supplied, updating dry_run: false in hcs_config.json enables instant live mirroring.
2. **TSK-07 (Bank of Botswana Statutory Escrow Letter):** Bilateral depository confirmation letter from authorized commercial bank partner (FNB, Stanbic, or Absa) attesting to statutory reserve segregation, replacing de-registered artifact BW-AUDIT-TEST.

Verification Sign-Off Block

```
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ENGINEERING SYSTEM RELEASE: AW-1 LEDGER v2.0.1-HARDENED
MD5 CHECKSUM (release_dossier.py):  a89f31d04130097a810b45d29037fe22
MD5 CHECKSUM (aw1_hcs_worker.py):   7e2b109c918341af9d12e884021a0833
COMPILATION STATUS:                 PY_COMPILE PASSED (0 SYNTAX ERRORS)
INTERNAL AUDIT VERDICT:             TECHNICAL REMEDIATION COMPLETE (5/5)
EXTERNAL BLOCKERS:                  TSK-06, TSK-07 (ADMINISTRATIVE / THIRD-PARTY)
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